

Building Resilience: The Stress Response as a Driving Force for Neuroplasticity and Adaptation

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ABSTRACT

People exhibit an extraordinary capacity to adjust to stressful situations. Here, we argue that the acute stress response is a major driving force behind this adaptive process. In addition to immediately freeing energy reserves, facilitating a rapid and robust neurocognitive response, and helping to reinstate homeostasis, the stress response also critically regulates neuroplasticity. Therefore, understanding the healthy acute stress response is crucial for understanding stress resilience—the maintenance or rapid recovery of mental health during and after times of adversity. Contemporary resilience research differentiates between resilience factors and resilience mechanisms. Resilience factors refer to a broad array of social, psychological, or biological variables that are stable but potentially malleable and predict resilient outcomes. In contrast, resilience mechanisms refer to proximate mechanisms activated during acute stress that enable individuals to effectively navigate immediate challenges. In this article, we review literature related to how neurotransmitter and hormonal changes during acute stress regulate the activation of resilience mechanisms. We integrate literature on the timing-dependent and neuromodulator-specific regulation of neurocognition, episodic memory, and behavioral and motivational control, highlighting the distinct and often synergistic roles of catecholamines (dopamine and norepinephrine) and glucocorticoids. We conclude that stress resilience is bolstered by improved future predictions and the success-based reinforcement of effective coping strategies during acute stress. The resulting generalized memories of success, controllability, and safety constitute beneficial plasticity that lastingly improves self-control under stress. Insight into such mechanisms of resilience is critical for the development of novel interventions focused on prevention rather than treatment of stress-related disorders.

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Prolonged exposure to stress has well-known negative consequences for both mental and physical health (1,2). However, humans also show a remarkable ability to adapt to stressful environmental changes and increase their capacity to cope with future stressors (3). It is increasingly acknowledged that the stress response, through a complex array of neuro-hormonal and neurotransmitter changes, is the major driver of this adaptation (4). This response mobilizes coping resources by rapidly freeing up energy reserves, changing neurocognitive functions to facilitate rapid and rigid responses to immediate threats, and helping to reinstate homeostasis (5). Critically, the stress response is also a major regulator of neuroplasticity (6)—the brain's ability to modify, change, and adapt both structure and function throughout life in response to either intrinsic or extrinsic experience (7,8). The stress response thus enables the reinforcement of those stress-coping strategies that have proven to be successful. Therefore, investigating the healthy stress response and its adaptive function is vital for understanding stress resilience.

Resilience has been defined as the maintenance or quick recovery of mental health during and after periods of adversity (9). According to this definition, resilience is an outcome of good long-term mental health in the face of stressor exposure

(10). Contemporary resilience research differentiates between resilience factors (RFs) (Figure 1A) and resilience mechanisms (RMs) (Figure 1B). RFs encompass a spectrum of social, psychological, or biological variables, such as stable social support, specific personality traits, coping strategies, optimistic appraisal biases, features of brain architecture and function, or genetic predispositions. These factors serve as predictors of resilient outcomes in individuals facing adversities. In contrast, RMs refer to the dynamic physiological, psychological, or behavioral responses that manifest when individuals are confronted with stressors. These responses facilitate an individual's capacity to navigate life's challenges effectively, fostering not only recovery from the current stressor but also building resilience against future adversities while preserving mental and physical well-being (10,11).

RFs are dispositions or conditions that make the activation of these RMs more likely (11). For example, psychological resilience research suggests that a positive appraisal style (RF) promotes a timely recovery from acute stress responses (RM), and this activation of an RM under stress mediates the effect of the RF on resilience (12–14). Importantly, while some RFs may be stable and hard wired (e.g., traits, genotype), others may be modifiable and therefore strengthened by experience or

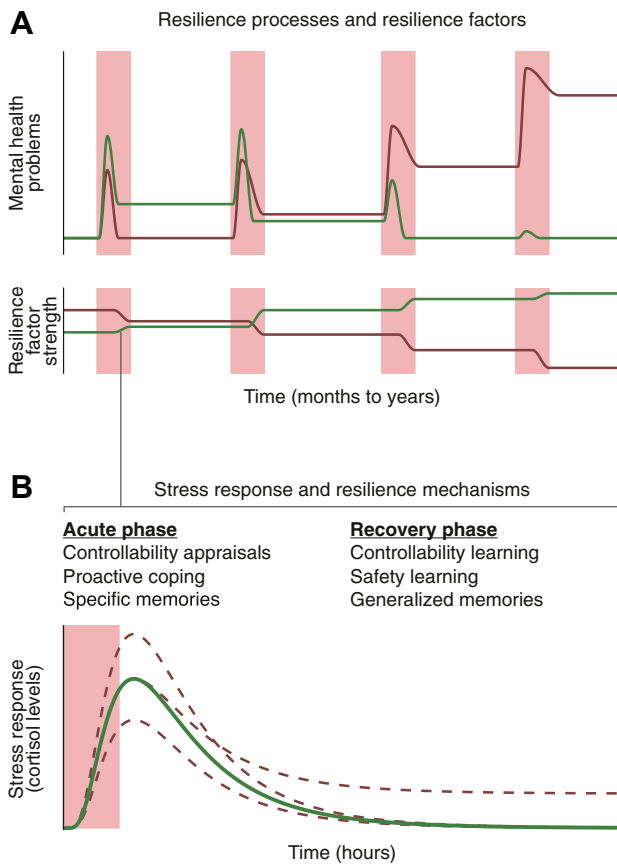


Figure 1. Illustration of the concepts of resilience factors, processes, and mechanisms. **(A)** Example of 2 trajectories of mental health problems in response to exposure to periods of prolonged stress (e.g., potentially traumatic events; indicated by red background bars) taking place at a time scale of months to years. The green lines in the 2 panels show a resilience process of someone who initially responds to adversity with a decline in mental health but who manages to strengthen a resilience factor (e.g., proactive coping strategies) and consequently exhibits a progressively more favorable response to repeated exposure (i.e., resilience). The red lines illustrate a trajectory of someone who exhibits a negative change in a resilience factor over time (e.g., declining positive appraisal style) and consequently shows progressively longer recovery periods and a trend toward worsening mental health. **(B)** Illustration of proximate resilience mechanisms (taking place on a time scale of hours) that are activated during the response to a stressor (red background bar) and are reinforced through stress-induced neuroplasticity. The green solid line depicts healthy activation of the cortisol stress response, an example of one of the stress effector systems involved in regulating the activation and plasticity of resilience mechanisms during the acute and recovery phases of the stress response. Dashed lines illustrate aberrations (hyperresponse, hyporesponse, or failure to return to baseline) that may compromise effective regulation of resilience mechanisms.

training. For example, successful active coping during stress (RM) could improve an individual's ability to deal with future stressors by boosting metacognitive representations of environmental controllability, resulting in heightened self-efficacy (RF). Experiences of relief and safety following stressful episodes can strengthen long-term resilience by bolstering neural circuits that underlie extinction learning. Finally, reappraisal under acute stress may foster a habit of interpreting negative situations more positively, thereby promoting a positive

appraisal style (RF) that supports long-term emotional resilience. Such processes of long-term change of RFs, referred to as resilience processes (Figure 1A), can explain the phenomenon of stress inoculation for example, whereby exposure to moderate and/or controllable stressors makes individuals more resilient to stress in the future, presumably because they have developed more efficient or new coping strategies (15,16).

There are several comprehensive reviews on biological (17,18) and psychosocial (19) RFs. Here, we focus on RMs, starting from the premise that a wide range of RFs ultimately converge on a limited set of key RMs that operate during exposure to stressors. As we will discuss, these RMs are tightly regulated by stress-related hormonal and neurotransmitter changes. We will develop the notion that stress resilience emerges from the utilization and success-based reinforcement of effective neurobiological, psychological, and behavioral coping actions in response to acute stressors. We propose that such processes induce beneficial plasticity—implicit and explicit memories of success, mastery, control, and safety—thereby leading to a long-term higher ability for efficient self-control under stress. More specifically, this plasticity-dependent resilience process results from a biphasic regulation consisting of 1) active (behavioral or cognitive) coping during the acute phase of the stress response and 2) the consequential creation or strengthening of a generalized safety and/or controllability memory during the recovery phase of the stress response. These phases occur under different stress-related neurochemical conditions, each relying on distinct neural circuits, and both are necessary for long-term adaptation and implicated in stress inoculation (16). Before discussing plasticity-driven resilience, we will summarize the effects of stress on neurocognition that allow for coping with immediate stressors.

THE STRESS RESPONSE AND IMMEDIATE COPING WITH STRESSORS

Stress is classically defined as a perturbation of our mental and physical equilibriums that results from unpredictable and/or uncontrollable adversity (20–23). However, the stress response is a complex and vital system that is crucial for surviving life's challenges (1). It has been studied in detail in animal models and is remarkably similar across many species (4). A vast literature has documented the cascade of neuroendocrine changes triggered during stress (24). At the cellular level, stress-sensitive neurotransmitters and hormones exert distinct modulatory effects on neural tissues. Such effects are targeted in both space and time and can thereby regulate adaptive neurocognitive functioning in a temporally dynamic fashion (25,26). Here, we will focus on catecholamines and glucocorticoids, but it should be noted that the stress response encompasses a broader array of neuropeptidergic, monoaminergic, and neuroendocrine changes (24).

The acute phase of the stress response is characterized by a surge of catecholaminergic activity, including rapid increases in both norepinephrine (27) and dopamine (25), as well as synergistic effects of glucocorticoids, mainly through mineralocorticoid receptors (28). These neurochemical changes result in heightened arousal, attentional reorienting to salient

events, and temporary suspension of higher-order cognitive processes required for long-term goal-directed behavior (5). This neurocognitive change has been explained by threat- and motivation-related activation of limbic and striatal circuits, as well as temporary suppression of learning and memory-related activation of the hippocampus and (parts of) the prefrontal cortex (PFC) (29). Neuroimaging research in humans has shown how these regional effects are part of comprehensive shifts in dominant configurations of large-scale neural systems. For example, acute stress results in activation of the salience network, including limbic and mesolimbic regions such as the amygdala, striatum, dorsal anterior cingulate cortex, anterior insula, temporoparietal junction, thalamus, and hypothalamus. These regions integrate information necessary for optimal responses to homeostatic threats, from sensory intake and appraisal to generating physiological and behavioral responses (30,31). Stress-related neurocognitive changes and activation of peripheral stress effector systems may thereby also mutually reinforce one another. At the same time, suppression is observed in default mode network-related regions, which include the (ventro)medial PFC (vmPFC) and the inferior parietal, posterior cingulate, and retrosplenial cortices, but also medial temporal lobe regions including the hippocampus (32). Consistent with work on PFC circuit dysfunction in animals, disruptions of functionality under acute stress are also observed in the executive control network, which includes the dorsolateral PFC, precentral and superior frontal sulci, dorso-medial PFC, and posterior parietal regions that support higher-order cognitive functions and goal-directed planning (33).

The restoration and recovery of homeostasis in the aftermath of acute stress also actively involves stress-sensitive hormones and neurotransmitters (1). Neuroparmacological studies implicate a role for slow effects of corticosteroids in this recovery phase, likely through genome transcription-dependent processes (24). Neurally, this phase is characterized by restored dominance of the executive control and default mode networks (5) and is thought to support consolidation, integration, and contextualization of novel memories (6). We will discuss below how during this key phase, experiences made during acute coping translate into longer-term changes that impact the way an individual will anticipate, process, and react to future stressors and explain how different RMs and protective neuroplastic states are activated during distinct phases of the stress response.

STRESS-RELATED NEUROPLASTICITY RELATED TO MEMORY PROCESSING

Contemporary theories of predictive coding emphasize that stressful episodes are generally experiences that violate an organism's predictions regarding its environment (34,35). Therefore, the rapid catecholaminergic changes in response to acute stressors described above can be conceptualized as prediction error signals. Both dopaminergic (36–38) and noradrenergic (39) activation have been attributed a role in signaling such (unsigned, or absolute) prediction errors. It is thought that the function of the stress response is to resolve uncertainty created by these prediction errors by generating attentional shifts, improving sensory signal-to-noise ratios, and regulating memory storage. Thus, these theories emphasize

that the stress response, in addition to gating sensory intake, is primarily a major driver of learning and neuroplasticity (40). All major stress-sensitive hormones and neurotransmitters, including norepinephrine (41), dopamine (42), and cortisol (28), have been linked to heightened neuroplasticity. Recently, it has become clear how these neuroplasticity-related stress effectors operating in different temporal domains also affect distinct aspects of learning and memory (6), thereby potentially initiating processes that enhance resilience.

Enhanced neuroplasticity during acute stress may strengthen memories of coping strategies that would be beneficial during a future encounter of a similar stressor. However, given that acute stress is also known to hamper memory retrieval (43), stress-induced neuroplasticity may also serve a role in improving future predictions (44). This would allow organisms to better anticipate potentially stressful situations and adopt strategies to avoid exposure, thereby minimizing future prediction errors and the stress responses that they trigger.

The effects of stress on neuroplasticity have been most prominently investigated in the realm of episodic memory, which pertains to the recollection of the time and place in which events occurred. A vast body of research, reviewed elsewhere (6), has shown how people acquire exceptionally strong episodic memories of stressful events. Recent findings are beginning to delineate how stress-sensitive neurotransmitters and hormones not only exert quantitative effects on memory strength but also contribute intricately to regulating specific qualities of memories acquired during stressful episodes. For example, animal models show that rapid norepinephrine activation particularly enhances accuracy of memories acquired during acute stress, primarily by rapidly enhancing hippocampal plasticity (45). This mechanism appears to allow for an unselective increase in storage capacity during phases of acute stress when it is not yet clear which information will prove to be relevant later. Noradrenergic mechanisms are also involved in the later selection of memories that are retained, not only in promoting synaptic consolidation during the recovery phase of the stress response but intriguingly also by selectively consolidating memory traces that were associated with more arousal already during (46) or potentially even after (47) encoding [but see (48)].

Recent work has shown that stress hormones also play an important role in the early phase of systems consolidation, the slow process by which memories undergo a reorganization and become gradually less dependent on the hippocampus and more integrated into neocortical networks. Such associative networks, or schemas (49), prominently involve ventral and medial parts of the PFC (50). The slow process of systems consolidation results in a gradual transformation of memories from specific recollections that are rich in episodic detail and context toward a more generalized, gist-like, or schema-based representation. Postlearning infusion of norepinephrine into the basolateral amygdala leads to a retention of hippocampus-dependent memory specificity, even during remote recall (45). Similarly, postencoding noradrenergic activation using the α_2 -adrenoceptor antagonist yohimbine led to a reduced decline of memory accuracy and increased hippocampal activation during memory recall in humans (51).

Intriguingly, studies also implicate a role for slower effects of cortisol in the recovery stage of the stress response that is quite distinct from norepinephrine; rather than enhancing episodic specificity, glucocorticoids appear to promote generalization of newly acquired memories beyond the original episodic spatiotemporal context. For example, in contrast to noradrenergic infusions, systemic glucocorticoids administered after learning induced a strong but generalized memory of the context in which rodents received shock (52). Furthermore, other studies have shown how postlearning manipulation of glucocorticoids results in generalization of ensembles of dentate gyrus engram cells (53) and that higher glucocorticoid levels during encoding are associated with an acceleration of memory generalization (54). Furthermore, this latter effect was blocked by administration of metyrapone, which inhibits the production of glucocorticoids (55), which suggests that glucocorticoids are causally involved in promoting mechanisms that underlie systems consolidation, thereby facilitating generalization (56). These findings show how stress hormones optimize predictions of future threats; while arousal-related noradrenergic activation enhances the representation of specific threats, stress levels of glucocorticoids promote a generalization process that relies more on cortical cognitive machinery, helping us to infer similarly dangerous future situations, although it should be noted that overgeneralization may result in maladaptive fear responses.

STRESS-RELATED NEUROPLASTICITY AND REINFORCEMENT OF BEHAVIORAL COPING STRATEGIES

Research has shown that the effects of stress hormones on hippocampal plasticity are mediated by the amygdala, which integrates neuromodulatory influences and modulates mnemonic activity and synaptic plasticity in other brain regions (57,58). Furthermore, more recent evidence shows that the effects of stress on mnemonic processes extend beyond episodic and hippocampus-dependent memory (6). For example, noradrenergic activation of the basolateral amygdala modulates functional interactions between the prelimbic cortex, insula, and hippocampus that are involved in gating effects of glucocorticoids on object recognition memory in rodents (59). It is plausible that such stress-induced hyperplasticity in regions throughout the brain is responsible for the reinforcement of specific types of appraisals and behaviors that are observed in response to stressor exposure.

A classic example of such plasticity is seen in the long-term consequences of exposure to uncontrollable stressors (60). Such consequences have been widely studied in experiments in which experimental animals or humans are exposed to either escapable (controllable) shocks or inescapable (uncontrollable) shocks that are yoked to the escapable shock. This creates a manipulation of controllability, a key dimension of stressor appraisal (10), while controlling for the amount of adversity (i.e., shocks) that is experienced. Research has shown that such experiences of uncontrollability, or failures to adequately respond to stressors, reinforce perceptions of helplessness (61), promote reactive and inflexible coping strategies (62), and impair working memory performance (63). Patients with stress-related disorders are also particularly susceptible to the

consequences of experiences of uncontrollability, which lead to a bias toward underestimation of the controllability of novel situations (64,65).

Recent work has shown that experiences of successful coping reinforce appraisals of controllability, manageability, and agency (10,66), as well as active and flexible coping mechanisms that support adaptive stress-response regulation. This leads to the effective deployment of RMs. A prominent example of this is seen in the literature on the stress inoculation hypothesis: the notion that limited exposure to intermittent and controllable stressors, often studied early in life, leads to improved resistance or resilience to stressors when tested later in life (16). For example, giving mice the opportunity to actively exert control over stressors results in lasting reduced stress susceptibility that generalizes across a variety of behavioral and physiological stress assays, notably including uncontrollable stressors (67,68). Likewise, successful active coping reduces reactive and inflexible responses during subsequent uncontrollable Pavlovian learning in both rats (69) and humans (70). Additionally, it can lead to increased detection of safety and consequential downregulation of stress responses that are no longer necessary, as reported in a study showing enhanced extinction of Pavlovian conditioned fear responses following experiences of controllability (71). Exposure to controllable stressors can also bias away from inflexible Pavlovian learning toward flexible, instrumental learning and intriguingly does so even beyond levels that have been observed in nonstressed animals (72). Thus, learning through experience about environmental controllability leads to a shift from negative controllability appraisals, pronounced stress reactivity, and inflexible, reactive coping styles to generally more positive controllability appraisals [or higher sense of agency (66)], less pronounced stress responses, and more flexible, proactive coping patterns. This shift may contribute to long-term resilience through optimized stress responses (66), provided that the individual does not overestimate environmental controllability (10).

The neural mechanisms that underlie these phenomena are becoming increasingly clear. Reactive and proactive behavioral coping strategies that are activated during the acute phase of the stress response have been shown to rely on distinct neural circuits (73,74). While reactive behaviors are implemented in amygdala-centered networks (58,75) and operate according to Pavlovian (action outcome-insensitive) principles, proactive behaviors engage (ventral) striatal regions and rely on instrumental learning (i.e., weighing consequences of previous actions) (76). Development of proactive coping strategies, for example in active avoidance paradigms in rodents, has been linked to a shift in activity from the amygdala to the ventral striatum (76). Rodent research has implicated the vmPFC and posterior dorsomedial striatum in detecting environmental controllability and switching between reactive and proactive stress-response strategies (72,77). While the human literature on uncertainty, reinforcement learning, and value-based decision making also implicates more anterior parts of the PFC (78), neuroimaging studies that have directly compared controllable versus uncontrollable aversive stimulation support the notion that the (vm)PFC plays an important role in representing controllability (79–82). Furthermore, animal studies have established that mice that show higher levels of resilience exhibit stronger phasic

dopaminergic activity in the ventral striatum (nucleus accumbens) when actively engaging a threatening conspecific (83). Notably, this dopaminergic activity was observed in a mesolimbic pathway that was shown within the same study to be specifically responsive to rewards and inhibited by aversive stimuli. These findings suggest that resilient mice process aspects of acutely stressful social situations similarly to rewards, which may in turn play a causal role in building their resilience to chronic social stress.

A prospective study with humans showed that increased ventral striatal responsiveness to rewarding relative to punishing outcomes in a gambling task predicted less posttraumatic stress disorder severity 1 year later, after exposure to stressful military service (84). In civilian trauma survivors who were followed prospectively, a preserved ventral striatal response and its connectivity to the vmPFC to a rewarding outcome from shortly after trauma to 1 year following exposure was also associated with less posttraumatic stress disorder severity (85). These studies indicate that the responsiveness of the reward system generally constitutes a likely RF, presumably facilitating the recognition of control during and after stress.

Experiences of (un)controllability have been shown to result in long-term adaptations in neurobiological pathways involved in detecting controllability and switching between reactive and proactive stress-response strategies, such as the vmPFC and posterior dorsomedial striatum. Research has shown that upon detection of controllability, these circuits exert inhibitory control not only over the amygdala but also over serotonergic neurons in the dorsal raphe nucleus (72,77). Long-term exposure to uncontrollable stressors results in progressive sensitization of the dorsal raphe nucleus (60), a process that involves corticotropin-releasing hormone (86). In contrast, exposure to controllable stressors results in a persistent capacity of this ventromedial prefrontal circuit to exert inhibitory control over this region (72). This process of long-term change requires NMDA receptor and ERK (extracellular signal-regulated kinase) signaling in the vmPFC (87) and is diminished following injection of a protein-synthesis inhibitor (88), demonstrating that this adaptive process depends on neuroplasticity in the vmPFC.

STRESS-RELATED NEUROPLASTICITY AND SAFETY LEARNING

Phasic dopamine activity in the nucleus accumbens is also necessary to encode rewarding experiences of relief and safety at the end of the acute phase of the stress response. For example, research with both rodents and humans has shown the involvement of dopamine in fear extinction learning (89–92), a process that is also known to depend on the vmPFC (93). Furthermore, fear extinction retention can be enhanced through coupling of rewards to previously threat-conditioned stimuli during extinction training or aversive-to-appetitive counterconditioning (94). Temporally more extended prefrontal release of dopamine during the recovery phase after stress (25) may help translate such experiences into long-term safety memories, as is suggested by studies that have shown beneficial effects of dopamine on the consolidation of extinction memories (95–99), a process that promotes lasting reduced fear responses, is dependent on the ventromedial prefrontal aspects of the reward system, and involves spontaneous vmPFC reactivations in the aftermath of

extinction learning (95,97). Notably, a recent study with mice showed that the magnitude of relief from physical stress predicted individual resilience to the development of a depressive phenotype following chronic stress (100). This relief signal was conveyed through the reward-related mesolimbic dopaminergic pathway. Blocking this pathway negatively affected resilience, whereas rewards presented at the offset of the stressor were shown to promote resilience. Although it is currently unclear whether mechanisms that underlie extinction learning and relief overlap, these findings together suggest a causal role for dopaminergic activity at the transition from the acute to the recovery phase of the stress response in promoting resilience (101).

Evidence concerning the role of dopaminergic pathways in safety learning is consistent with studies that have shown beneficial effects of glucocorticoids on extinction (28,102,103). For example, administration of hydrocortisone (i.e., a synthetic form of endogenous cortisol) before extinction reduces expression of fear memory during extinction training and increases vmPFC connectivity to the amygdala, resulting in downregulation of the amygdala-hippocampal complex (104). Glucocorticoid activation during extinction learning also makes extinction memories less context dependent, showing generalization of safety learning (105,106). This finding is consistent with studies showing that glucocorticoids also facilitate the generalization of episodic-like memories across different contexts (56), which were mentioned above. Notably, although this remains to be investigated directly, safety learning generalization may constitute a meta-learning process by which organisms gradually acquire a representation of the controllability of their surrounding environments in the aftermath of stress. Thus, these notions are consistent with the literature reviewed above on the role of the vmPFC in detecting controllability and regulating serotonergic neurons in the dorsal raphe nucleus and suggest a broader role for the vmPFC in estimating agency and generalizing experiences into broad controllability representations that may to some extent be immune to violations of objective controllability.

As also mentioned above, effects of glucocorticoids may extend toward dorsal regions of the PFC that support higher-order cognitive functions. For example, exposure to stress in rats triggers an enhancement of glutamatergic transmission in pyramidal neurons in the PFC that is consistent with genomic corticosteroid effects (107). Oral administration of the synthetic glucocorticoid hydrocortisone has also been shown to lead to a delayed enhancement of PFC function and working memory accuracy in humans (108,109). Studies have shown beneficial delayed effects of stress induction and hydrocortisone on cognitive regulation of emotion in tasks that require reappraisal strategies (110,111).

Together, these findings identify the acute response to stressors as the major driver of adaptation to adverse environmental circumstances and point toward a dual window for activating and reinforcing RMs. The first window occurs during the acute phase of the stress response, when experiences of successful coping would lead to appraisals of controllability and a more effective shift away from the amygdala and toward the ventral striatum upon subsequent encounters with the same or similar stressors. The second window occurs during the recovery phase, during which activation of the (vm)PFC and hippocampus would support the integration of safety experiences into generalized long-term memories.

CONCLUSIONS AND IMPLICATIONS

In this perspective article, we reviewed emerging literature on stress-induced activation of RMs, which we defined as the proximate mechanisms that drive long-term changes in RFs. While negative changes in RFs may culminate in maladaptation and psychopathology, we highlighted resilience processes that enhance resilience in response to future stressors. We started from the premise that the stress response, through intricate neurochemical changes, is the primary driver of adaptation and neuroplasticity. We reviewed and integrated literature on the effects of stress-related neuromodulators on neurocognition (5) and memory (6) with research on appraisal and behavioral coping strategies to identify synergies that foster understanding of stress-induced neuroplasticity that reinforces beneficial appraisal and coping strategies. We identified several synergies, such as catecholaminergic pathways reinforcing proactive behaviors to cope with stressors and mechanisms that underlie the generalization of episodic memories. These insights shed light on how people acquire generalized beliefs of environmental controllability, safety, and agency.

One challenge that remains is integrating animal and human research on resilience. While animal models inherently lack the broader psychological, sociocognitive, and cultural dimensions that are central to human stress resilience, they are critically important for investigating biological RFs and RMs. Aligning the operationalization and measurement of resilience across species is essential to optimize the validity of these models. For example, emerging methods for high-resolution behavioral monitoring are making it possible to quantify naturalistic behavior in animals (112), thereby enabling the use of longitudinal designs and behavioral outcome measures that more closely match those used in human resilience research. Such improvements would also allow us to harness novel neurotechnologies in animal research that are too invasive for use in humans, which would allow for causal inferences regarding the neural circuits involved.

Insights into RMs during stress carry important clinical implications. Key RFs that determine an individual's ability to adapt to stressful situations form an acquired neurobehavioral disposition that activates specific RMs during stress. This implies that resilience cannot be explained solely by the absence of vulnerability factors. However, due to the historical emphasis on neurobiological disease mechanisms and vulnerability factors, an understanding of the neurobiological, psychological, behavioral, and social processes that enhance resilience is just beginning to emerge. This is critical because despite substantial progress in understanding brain mechanisms of mental disease, the burden of stress-related disorders has not declined (9). Consequently, there is a shift from focusing on the treatment gap to emphasizing the need to close the prevention gap (113). Refocusing research efforts on mechanisms of resilience is consistent with this agenda.

The critical next challenge facing this field is understanding how to bolster RMs during stress. In our conceptualization, effects of multiple RFs converge onto a limited set of RMs. The fact that some RFs are malleable implies that preventive interventions to strengthen resilience can utilize multiple entry points, targeting social, behavioral, or psychological factors or

even directly targeting biological RFs. Insights into proximate RMs that operate during stress will enable the optimization of interventions targeting RFs as well as assessment of whether particular interventions do in fact target RMs of interest. Another possibility is to directly intervene in the operation of RMs. Examples of such interventions include application of cognitive techniques [e.g., training in cognitive reappraisal (114) or reward sensitivity (115)] or even direct neurobiological manipulations [e.g., acquisition of self-control over neural activation during stress using targeted neurofeedback (116,117)]. Notably, both could be directly linked to stressful experiences in real life using just-in-time adaptive interventions that train individuals to apply context-appropriate strategies at moments when they are needed (118).

In conclusion, we reviewed literature converging on the notion that stress resilience is strengthened through the use and success-based reinforcement of effective coping strategies during acute stress, thereby creating optimally generalized memories of success, controllability, and safety. Understanding such beneficial plasticity that enhances self-control under stress is crucial for developing novel interventions that aim to prevent rather than treat stress-related disorders.

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